MIL-V-22133C(SHIPS)
26 May 1969
SUPERSEDING
MIL-V-22133B(SHIPS)
20 March 1961
(See 6.5)

MILITARY SPECIFICATION

VALVES, RESILIENT AND METAL SEATED, BUTTERFLY

WORKING PRESSURE UP TO 200 PSI, 180°F. MAXIMUM

(SHIPBOARD USE)

1. SCOPE

1.1. Scope. This specification covers butterfly valves with working pressures up to 200 pounds per square inch (psi), 180°F. temperature for use on shipboard.

1.2 Classification. Butterfly valves shall be of the following types, classes and series, as specified (see 6.2):

Type I - Metal disc to non-metal seat
Type II - Metal disc to metal seat

Class A - Fresh and sea water
Series 50 - 50 psi maximum working pressure (type I only) (24 inches iron pipe size (ips) and above).
Series 200 - 200 psi maximum working pressure (20 inches ips and below).

Class B - JP-5 fuel.
Series 200 - 200 psi maximum working pressure (20 inches ips and below).

Class C - Oil, fuel (Navy special).
Series 200 - 200 psi maximum working pressure (20 inches ips and below).

Class D - Oil; diesel or lube.
Series 200- 200 psi maximum working pressure (20 inches ips and below).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-N-286 - Nickel-Copper-Aluminum Alloy, Wrought.
QQ-N-288 - Nickel-Copper-Alloy and Nickel-Copper-Silicon Alloy Castings.

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-R-196 - Repair Parts for Internal Combustion Engines, Packaging of.
MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-D-1000- Drawings, Engineering and Associated Lists.
MIL-P-5516- Packing, Preformed, Petroleum Hydraulic Fluid Resistant 160°F.
MIL-T-5624- Turbine Fuel, Aviation, Grades JP-4 and JP-5.
MIL-P-15137-Provisioning and Technical Documentation for Repair Parts for Electrical and Mechanical Equipment (Naval Shipboard Use).
MIL-B-23921-Bronze, Nickel Aluminum Castings for Seawater Service.

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STANDARDS

MILITARY

MIL-STD-100 - Engineering Drawing Practices.
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-167 - Mechanical Vibrations of Shipboard Equipment.

DRAWINGS

MILITARY

810-1385892 - Flanges, Bronze, Special For Use With Butterfly Valve.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A47 - Malleable Iron Castings.
A216 - Carbon-Steel Castings Suitable for Fusion Welding for High-Temperature Service.
A395 - Cast Ductile Iron for Pressure Containing Parts for Use at Elevated Temperatures.
A536 - Ductile Iron Castings.
B21 - Naval Brass Rod, Bar, and Shapes.
B61 - Steam or Valve Bronze Castings.
B143 - Tin Bronze and Leaded Tin Bronze Sand Castings.
B164 - Nickel-Copper Alloy Rod and Bar.
B438 - Copper-Base Sintered Metal Powder Bearings (Oil Impregnated).
B439 - Iron-Base Sintered Metal Powder Bearings (Oil Impregnated).

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

UNITED STATES OF AMERICA STANDARDS INSTITUTE (USAS)

B16.5 - Steel Pipe Flanges and Flanged Fittings.

(Application for copies should be addressed to the United States of America Standards Institute, 10 East 40th Street, New York, N.Y. 10016).

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1. Qualification. The type I, series 200 valves furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at time set for opening of bids (see 4.2 and 6.3).

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Table I - Materials

Name of Part	Type I					
	Classes A and B			Classes C and D		
	Material	Applicable documents	Remarks	Material	Applicable documents	Remarks
Body	Valve Bronze	ASTM B61		Ductile iron	ASTM A395 ASTM A536	
	Gun metal	ASTM B143	Alloy 1B	Malleable iron	ASTM A47	
	Nickel aluminum bronze	MIL-B-23921		Cast steel	ASTM A216	
Disc	Nickel copper alloy	ASTM B164 class A or QQ-N-288		Nickel copper alloy	ASTM B164 class A or QQ-N-288	
	Nickel aluminum bronze	MIL-B-23921	For other than sea-water	Aluminum bronze	MIL-B-23921	
	Valve bronze	ASTM B61		Valve bronze	ASTM B61	
				Corrosion resisting steel 13 Cr	Commercial	
Stem	Nickel copper alloy Nickel copper aluminum	ASTM B164, class A QQ-N-286		Nickel copper alloy Nickel copper aluminum	ASTM B164 class A QQ-N-286 Commercial	
	Corrosion resisting steel	Commercial	For other than sea-water As approved	Corrosion resisting steel		As approved
"O" ring	Elastomer	MIL-P-5516		Elastomer	MIL-P-5516	
"V" ring	Elastomer		As approved	Elastomer		As approved
Stem bushing	Bearing bronze	ASTM B202, type I	Oil impregnated	Bearing bronze	ASTM B202 type I	Oil impregnated
	Valve bronze	ASTM B61		Valve bronze	ASTM B61	
	Nickel copper alloy	ASTM B164 class A Commercial		Nickel copper alloy	ASTM B164 class A Commercial	
	Plastic bearing		As approved	Plastic bearing		As approved
Liner 1/	Elastomer		As approved	Elastomer		As approved
"O" ring	Buna N			Buna N		
Key Cap screws	K Monel	QQ-N-286	As approved	K Monel	QQ-N-286	As approved
	Nickel copper alloy	ASTM B164		Brass	ASTM B21 alloy A half hard	

See footnote at end of table.

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Table I - Materials (Cont'd)

Name of part	Type I					
	Classes A and B			Classes C and D		
	Material	Applicable document	Remarks	Material	Applicable document	Remarks
Backing ring for liner	Non-metallic		As approved	Non-metallic		As approved
	Valve bronze	ASTM B61		Carbon steel		As approved
	Carbon steel		As approved 2/			
	Type II					
All valve parts	Same as type I for each class except that aluminum bronze valve bodies are not permitted.					

1/ The liner and "O" ring material shall be suitable for the intended service. Liner and "O" ring material of production valves shall be same as that furnished in valves that have been qualified under this specification.
2/ Will be considered providing the backing ring is totally encapsulated in the liner.

3.3 Construction (see 6.2). Valves shall be of ample proportions to withstand the pressure of the system as well as the external forces transmitted to the valve through pipe and ship movements.

3.4 Design. The valves shall be of a design which shall withstand repeated operation without excessive wear on working parts.

3.4.1 Stops. The valves shall be provided with mechanical stops to prevent overtravel of the disc in open or closed position. Means shall be provided for positively locking the disc in the fully open and fully closed position as specified (see 6.2). Unless otherwise specified (see 6.2), all valves in sizes 8 inches and above shall be furnished with gear or lead-screw operators.

3.4.2 Operators. Operators shall be of material and design suitable for the specific valve application. The use of cast iron is not permitted. The valves shall be easily and quickly operable by one man without the use of mechanical aids other than those incorporated in the valve as specified (see 6.2). The maximum force required to seat or unseat the valve shall not exceed the values listed in table II.

Table II - Maximum allowable tangential force to seat valve based on valve handwheel or handle size.

Handwheel diameter	Handle length	Total tangential force on rim of handwheel or end of handle
Inches	Inches	Pounds
Below 5	---	30
5	---	50
6	4	72
7	4-1/2	90
8	5	102
9	5-1/2	114
10	6	120
11	6-1/2	129
12	7	135
14	8	138
16	9	141
18	10	144
21	11-1/2	147
24 and above	13 and above	150

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3.4.2.1 Remote operation. The valve shall be so designed that remote operating devices may be adapted thereto if required (see 6.2).

3.4.2.2. The valves shall be designed so that they close when the handle or handwheel is turned clockwise. Each valve shall be equipped with an indicator to indicate the position of the disc.

3.4.3 Body construction

3.4.3.1 Type I, class A, series 50 valves - Type I, class A, series 50 valves (24 inches ips and above), shall have flanged ends and the ends of the liner shall extend out over or beyond the flange to form a gasket. Unless otherwise specified in the contract or order, series 50 valves shall be furnished drilled. Flange dimensions and drilling shall be in accordance with table III.

Table III - Flange dimensions

Size	O.D.	Thickness	No. holes	Drilling 1/ Hole diameter P.C.	
Inches	Inches	Inches		Inches	Inches
24	32	1	20	1-3/8	29-1/2
30	38-3/4	1-1/8	28	1-3/8	36
36	46	1-1/4	32	1-5/8	42-3/4
42	53	1-3/8	36	1-5/8	49-1/2
44	55	1-3/8	38	1-5/8	51-1/2
48	59-1/2	1-1/2	44	1-5/8	56

1/ For valves having bonnet or bushing housing body configurations that interfere with the flange drilling, the bolt holes in way of the bonnet or bushing housing may be drilled or tapped.

3.4.3.2 Type I and II, classes A, B, C and D, series 200. Type I and II, classes A, B, C and D, series 200 valve bodies shall be of one of the flangeless constructions specified in 3.4.3.2.1 or 3.4.3.2.2 or a combination thereof. There shall be a minimum of 2 centering holes included in the design of valves 14 inches to 20 inches inclusive.

3.4.3.2.1 Construction A. Construction A shall be of generally circular shape as shown on figure 1. The outside diameter of the valve body shall be equal to the specified bolt circle diameter of 150 pound flanges in accordance with USAS B16.5 minus the specified bolt hole diameter. Tolerances shall be in accordance with table IV.

Table IV - Tolerances.

Size of valve	Tolerance
Inches	
2 to 6	+1/32 -1/16
8 to 12	+1/32 -3/32
14 and above	+1/32 -1/8

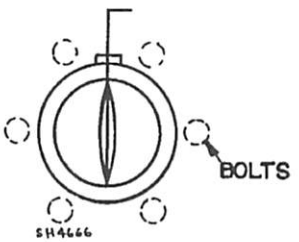


Figure 1 - Construction A.

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3.4.3.2.2 Construction B. Construction B shall be scalloped around the outer periphery with the holes drilled through the lugs as shown on figure 2. The drilling shall be in accordance with 150 pound flanges specified in USAS B16.5. Lugs may also be furnished drilled and tapped, National Coarse, when specified (see 6.1.6.1 and 6.2).

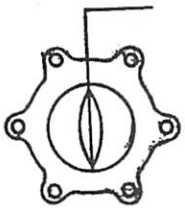


Figure 2 - Construction B.

3.4.4 Seats.

3.4.4.1 Type I. Seat for type I valves shall be metal to non-metal. The non-metallic member shall be in the form of a liner.

3.4.4.1.1 Liner, type I. The liner shall extend partially over or beyond the flange face or body face. It shall be so designed that when the mating flanges are pulled up, the liner shall be compressed, thus acting as line gaskets. Permanent bonding or use of adhesives between body and rubber liner is not permitted. The method of securing the liner or sleeve in all series valves shall be such that the liner or sleeve can be easily renewed without the use of special tools or devices. Body and sleeve construction shall be such that under a vacuum of 20 inches (mercury), the sleeve shall not collapse or obstruct the waterway of the valve. The liner shall be so designed that under rated service conditions it shall not balloon or deform in any manner that will be detrimental to normal valve operation.

3.4.4.2 Type II. Seats for type II valves shall be metal-to-metal (no non-metallic liner) and shall be suitable for severe throttling service. Metal seats shall be a part of the body. Type II valves are not intended to be used for tight shut-off.

3.4.4.2.1 Leakage rate. Leakage rate of type II valves shall not exceed 8 gals/min/inch of ips at rated pressure.

3.4.4.2.2 Stem seal. Stem seal shall be designed in a manner that will eliminate external leakage of the stem.

3.4.4.2.3 Line seal. Valve faces shall be serrated finish.

3.4.5 Valve dimension. Valve (face-to-face) dimensions shall be in accordance with table V. These dimensions apply to the (face-to-face) dimensions of the liner in the uncompressed condition. Where construction A valves (see 3.4.3.2.1) are manufactured the body outside diameter shall be in accordance with table V.

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Table V - Valve dimensions.

Valve size	Diameter of waterway	Face-to face type I	Face to face type II	O.D. of body construction A valves
Inches	Inches	1/ Inches		
		SERIES 200		
2	The diameter of	1-3/4	1-5/8	4
2-1/2	the valves size	1-7/8	1-3/4	4-3/4
3	2 through 12 inches	1-7/8	1-3/4	5-1/4
3-1/2	shall be equal	2	1-7/8	6-1/4
4	to the inside	2-1/8	2	6-3/4
5	diameters of the	2-1/4	2-1/8	7-5/8
6	corresponding sizes	2-1/4	2-1/8	8-5/8
8	of USAS schedule 40	2-1/2	2-3/8	10-7/8
10	pipe.	2-13/16	2-11/16	13-1/4
12		3-3/16	3-1/16	16
14	13-1/4	3-3/4	3-5/8	17-5/8
16	15-1/4	4-1/8	4	20-1/8
18	17-1/4	4-5/8	4-1/2	21-1/2
20	19-1/4	5-1/8	5	24-1/4
		SERIES 50		
24	23-1/4	6-3/16		
30	29-1/4	6-23/32		
36	35-1/4	8-1/8		
42	41-1/4	10-1/8		
44	43-1/4	10-5/8		
48	47-1/4	11-1/8		

1/ A tolerance of plus or minus 1/16 inches shall be permitted on the face-to-face dimension of valve sizes 2 through 6 inches; plus or minus 1/8 inch for sizes 8 through 20 inches and plus or minus 3/16 inches on size 24 inches and above.

3.4.6 Valve stems. Valve stems shall be either of one piece construction extending through the valve disc or of a construction comprising 2 separate stub shafts inserted into the top and bottom disc hubs. The cross sectional area of the stem shall be adequate for operation of the valve under all service conditions.

3.4.7 Disc and stem assembly. Disc and stem assembly shall be such that torque is transmitted by the shape of the stem or by keys or cap screws. The size and number of the keys or cap screws and the cross sectional area of the stem in way of the keys or cap screws shall be adequate for operation of the valve under all service conditions. Where the disc is secured to the stem by means which penetrate both disc and stem, O-rings or Welch plugs shall be provided to prevent leakage between the stem and disc. The use of taper pins to secure the valve disc to the shaft is prohibited.

3.4.7.1 Cap screws. When cap screws are used, the stem shall be drilled and the ends of the screws shall extend through the stem into the other side of the disc. The hole in the disc engaging the end of the cap screw shall not be drilled through. The portion of the screw which penetrates the disc and stem (two-places) shall not be threaded.

3.4.8 Packing and stuffing box. The valve stems shall be sealed against leakage by using O-rings or stuffing boxes with square or "V" type packing.

3.4.9 Shock and vibration resistance. All valves furnished under this specification shall be of a design which has successfully passed shock tests in accordance with MIL-S-901 and vibration tests in accordance with MIL-STD-167.

3.4.10 Marking. Each valve shall have the following data distinctly cast or stamped on the body in a way that is not detrimental to the valve or stamped on an identification plate which shall be permanently secured to the body in a way that is not detrimental to the valve.

(a) Size
(b) Specification number (MIL-V-22133)
(c) Pressure rating:
"50" (for series 50 valves)
"200" (for series 200 valves)
(d) Class
(e) Name, initials or trademark of the manufacturer.

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3.5 Drawings.

3.5.1 Preliminary drawings. Within 30 days after award of contract, the contractor shall submit to the procuring activity preliminary drawings that are sufficient to permit evaluation of the design and approval of materials. It is preferred that the drawings be in accordance with 3.5.2. These drawings shall be detailed assembly drawings and shall show the following:

(a) A sectional assembly of the valve and details of the seat disc and stem assembly (see 3.4.7).
(b) Bill of material with specifications, grade and condition listed or other data adequate to identify the material proposed.
(c) Dimensions. - Overall, accessibility space including disassembly, clearances and all other dimensions pertinent to installation.
(d) Notes. - Design and performance characteristics, operating torque, weight and any limitations on installation.

3.5.1.1. Approved drawings. If the contractor proposes to furnish equipment which previously has been supplied and for which the procuring activity has approved drawings on record, copies of the approved drawings shall be acceptable in lieu of submitting preliminary drawings.

3.5.2 Production drawings. When specified (see 6.2), detail assembly drawings shall be submitted to the procuring activity by the contractor for completion of the contract or order. These drawings shall be detail assembly drawings as defined in MIL-STD-100 in general accordance with category F, form 3 of MIL-D-1000.

3.6 Repair parts.

3.6.1 Onboard. When specified (see 6.2), onboard repair parts shall be furnished as specified in table VI.

Table VI - Repair parts.

Item	Quantities
Stem packing and valve liner	10 percent of ship's order but not less than one set for each class, series and size furnished.
Valve stem, disc and stem bushing	5 percent of ship's order but not less than one set for each class, series and size furnished

3.6.1.1 Parts list. Repair parts shall be listed in accordance with MIL-P-15137.

3.7 Workmanship. Valves shall be free from defects that affect their appearance or their operation. Castings shall be clean, sound and free from blow holes, hard spots, porosity, cracks and other injurious defects. They shall be smooth and well cleaned both inside and outside, and all fins and roughness shall be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Qualification tests. 1/ Qualification tests shall be conducted at a laboratory satisfactory to the Naval Ship Engineering Center. Qualification tests shall consist of the tests specified in 4.2.1 through 4.2.3. Test equipment shall be satisfactory to NAVSEC.

1/ Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.3 and 6.3.1).

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4.2.1 Two sample type I, class A, series 200 valves in the 6 inch size with 2 extra liners shall be submitted for qualification tests specified in 4.2.1.2 through 4.2.1.8. The material of the valves furnished shall conform to that of type I, class A valves. Tests shall be conducted in the order listed.

4.2.1.1 Type I, class A, series 200 valves which have successfully passed the tests specified in 4.2.1 shall also qualify all other classes and all sizes of valves of the same design, provided that valves above 6 inches (including operators) pass the test specified in 4.2.3 and proper materials are furnished and non-metallic parts of the valve successfully pass the test specified in 4.2.2.

4.2.1.1.1 In the event a manufacturer desires to run qualification tests on other than type I, class A, series 200 valves, the class submitted shall be subjected to the tests specified in 4.2. Approval shall be granted only for the class submitted.

4.2.1.2 Hydrostatic test. Hydrostatic tests shall be conducted on each series 200 valve body for strength to 300 psi with the disc open. Valves shall be completely assembled for this test. Duration of the test is 3 minutes. Leakage, sweating or visible deformation at any point shall be cause for rejection. The valve shall be cycled (opened and closed) for 100 times prior to hydrostatic testing for seat tightness.

4.2.1.2.1 Seat tightness. The valve shall be pressurized alternately on each side of the disc with water and examined for seat tightness. The differential test pressure shall be equal to 220 psi. The duration of the test shall be 3 minutes for each side. Any leakage shall be cause for rejection.

4.2.1.3 Torque test. A torque test shall be conducted to measure the force required for tight seating (no leakage permitted) at the test pressure and to unseat the disc. The amount of tangential force required to seat or unseat the disc shall not exceed the values specified in 3.4.2, table II.

4.2.1.4 Throttling test. The valve shall be subjected to a throttling test to determine liner suitability. The valve shall be equipped with a hand lever so designed to allow not more than 1/4 of 1 degree play before and after the test. Test procedure shall be as follows:

(a) Test shall be conducted with a water flow rate sufficient to produce a pressure drop of 75 ± 5 psi.
(b) Valve disc shall be set in a position of 25 degrees open from the closed position and locked in place.
(c) The test shall be conducted for a period of 24 hours which may be divided into as many as 5 tests of about equal duration. After the test is completed the following check shall be made:

(1) Seat tightness test in accordance with 4.2.1.2.1.
(2) Torque test in accordance with 4.2.1.3.
(3) Visual examination to insure that no appreciable loosening of the stem to disc connection and no erosion of the disc or liner has occurred.

(d) Tested valve and liner shall be used as one of the valves required for 4.2.1.5.

4.2.1.5 Heat exposure test. The heat exposure tests shall be conducted in an oven of sufficient capacity and range of temperature to meet the requirements specified herein.

4.2.1.5.1 Test assembly. The valves shall be assembled in a line in accordance with figure 3. The pipe line flanges shall be in accordance with USAS B16.5. The complete assembly shall be connected to the water supply and pump in such a manner that the assembly may be removed from the oven while hot without damage to the valves. Water pressure at 100 psi shall be maintained in the valve assembly, through the open valve (valve "B") and up to the disc of the closed valve (valve "A") during the entire heat exposure.

4.2.1.5.2 Method of preheating oven. The oven shall be preheated in such a manner that when the valve assembly is placed in the oven, the valve flange metal temperature shall rise to 400°F. in less than 30 minutes. The oven arrangement should be such that after 400°F. temperature is reached, the temperature can be maintained in the 400° to 500°F. temperature range for 20 minutes.

4.2.1.5.2.1 Steel plates. Steel plates of sufficient quantity for the purpose, may be place in the oven to increase its heat capacity. The steel plates can be heated to approximately 750°F. prior to placing the valve assembly in the oven. When the valve assembly is placed in the oven on the steel plates, the temperature of the steel shall transfer to the valve assembly bringing the valve assembly temperature to 400°F.

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4.2.1.5.3 Method of preheating valve assembly. The valve assembly pressurized to 100 psi as specified in 4.2.1.8 shall be placed in the preheated oven on the steel plates and the valve flange metal brought up to 400°F. within 30 minutes.

4.2.1.5.4 Method of heating valve assembly. When the valve assembly has been preheated to 400°F., the temperature of the oven shall be maintained at the point where the upper part of the temperature range (near 500°F.) is reached during the next 30 minutes. After 30 minutes, the valve assembly shall be removed from the oven and placed on a steel plate for rapid cooling.

4.2.1.5.5 Seat tightness and torque test. When the valve assembly has cooled to ambient temperature, the rate of seat leakage at 220 psi through the closed valve (valve "A") shall be checked, first in one direction and then in the reversed direction. With 220 psi on one side of the disc, the amount of force required to close the open valve (valve "B") shall be measured. The rate of seat leakage at 220 psi through valve "B" shall be measured in the same manner as valve "A".

4.2.1.5.6 Leakage rate. After the leakage and torque tests and following the heat exposure, each valve shall be operated, (open and close), 10 times and then each valve shall be retested for seat leakage and torque requirement as outlined in 4.2.1.6. The rate of seat leakage shall not exceed 20 gallons per minute (gpm) and the amount of force required to seat or unseat the disc while subjected to 220 psi shall not exceed the value specified in 3.4.2, table II.

4.2.1.6 Operating cycle endurance test. A new valve or a new liner may be used for the following test. Water at 220 psi shall be passed through the valve and the valve shall be fully cycled 2000 times at a rate of 2 to 6 cycles per minute (cpm). The valve shall be opened against full differential pressure only. During this cycling period, the following tests shall be conducted:

(a) After every 500 cycles, the valve shall be tested for seat tightness in accordance with 4.2.1.2.1.
(b) After every 500 cycles, the force required to open the valve shall be recorded. These forces shall not exceed those listed in 3.4.2, table II.

4.2.1.7 Mechanical and visual examination. Upon completion of tests specified in 4.2.1.2 through 4.2.1.6, the valves shall be completely disassembled and all parts checked for distortion or other damage. The valve parts, with the exception of the liner and packing, shall not be distorted or damaged.

4.2.1.8 Reports and test samples. Upon completion of the mechanical and visual examinations specified in 4.2.1.7 at least 5 reports of the tests and results validated by a NAVSEC representative and the sample liners used in the tests shall be forwarded to the NAVSEC.

4.2.2 Test for non-metallic parts.

4.2.2.1 Liner soak test. Liner soak test shall be conducted using JP-5 fuel in accordance with MIL-T-5624 or diesel oil. Each valve shall be installed in operating position in a pipe line with a minimum of 12 inches of pipe line on either side of the valve. The pipe line flanges for steel or iron body valves shall be either socket-weld or weldneck 150# schedule 40 flanges, in accordance with USAS B16.5-1961 standards. The pipe line flanges for non-ferrous body valves shall be in accordance with Drawing 810-1385892. The line and valve shall be completely filled with the specified test fluid with the test fluid maintained at 150°F., the valve shall stand for 48 hours with the disc open, and 48 hours with the disc closed.

4.2.2.1.1 Operating cycle endurance tests. The operating cycle endurance tests shall be conducted in accordance with 4.2.1.6 except that the fluid used for sleeve soaking (see 4.2.2.1) shall be used for the cycle test.

4.2.2.1.2 Steaming out test. Valves shall be subjected to steam at 212°F. for 8 hours with the disc in the open position. After the steaming out and the valve allowed to cool off to ambient temperature, the valves shall be cycled for 100 times (opening and closing) and the amount of torque required to seat and unseat the disc shall be measured and shall not exceed values listed in 3.4.2 (table II).

4.2.2.1.3 Seat tightness tests. Valves shall be hydrostatically tested with diesel fuel oil or JP-5 fuel to 220 psi alternately on each side of the disc with the opposite side open for a duration of 30 minutes. The valves shall not leak or sweat through or around the disc. The liner shall not show evidence of ballooning or bulging caused by the presence of test fluid between the sleeve and the interior of the valve body.

4.2.2.1.4 Mechanical and visual examination. Upon completion of test specified in 4.2.2.1 through 4.2.2.1.3, the valves shall be completely disassembled and all parts checked for wear, distortion or other damage. The valve parts shall not be excessively worn or be damaged in any way.

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4.2.3 Shock and vibration test. Valves shall be subjected to shock test in accordance with MIL-S-901 and vibration test in accordance with MIL-STD-167.

4.2.3.1 Conditions for conducting shock and vibration tests. Series 200, class A valves shall be tested while subjected to an internal pressure of 200 psi, series 50, class A shall be tested while subjected to an internal pressure of 50 psi. Valves shall be tested in both the open and closed position. Valves in the open position shall be pressurized throughout, valves in a closed position shall be pressurized on one side of the disc.

4.3 Material certification. The inspector shall require certification based on certifiable data that the valve parts conform to their applicable material specification.

4.4 Sampling for quality conformance inspection.

4.4.1 Lot. All valves of the same material, pressure rating, size and construction offered for delivery at the same time shall be considered a lot for purposes of quality conformance inspection.

4.4.2 Sampling for examination. A random sample of valves shall be selected from each lot offered for the examination specified in 4.5 with lot acceptance based on the sampling requirements of MIL-STD-105 at inspection level II, acceptable quality level (AQL) equals 1.0 percent defective.

4.4.3 Sampling for quality conformance tests. A random sample of valves shall be selected from each lot offered for the tests specified in 4.6 with acceptance based on the sampling requirements of MIL-STD-105 at inspection level III, AQL equals 1.0 percent defective.

4.5 Examination. Each of the sample valves selected in accordance with 4.4.2 shall be examined to verify compliance with all the requirements of this specification not involving tests.

4.6 Quality conformance tests.

4.6.1 Hydrostatic tests. Each sample valve selected in accordance with 4.4.3 shall be hydrostatically tested to the pressures indicated in table VII after cycling 100 times (opened and closed).

Table VII - Hydrostatic pressures

Position	Series 50	Series 200
	Psi	Psi
With valve in opened position; for strength and porosity.	75	300
With valve in closed position for seat tightness. Pressure shall be applied alternately in each direction with the opposite end open	55	220

4.6.1.1 Strength and porosity. The valves shall be tested for strength and porosity with the valve open. Pressure shall be applied for a minimum of 5 minutes. Leakage, sweating or visible deformation at any point on the surface shall cause the valve to be rejected.

4.6.1.2 Tightness. For tightness on seat with the disc closed by hand without exceeding the tangential forces listed in table II, the pressure to be applied for 3 minutes alternately on both sides of the disc with the opposite end open for examination in each case. Under this test the valve shall not leak water or sweat through or around the disc. Evidence of ballooning of the sleeve during this test shall be cause for rejection.

4.7 Inspection of preparation for delivery. The packaging, packing and marking shall be inspected for compliance with section 5 of this document.

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5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.4).

5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts.

5.1.1 Valves.

5.1.1.1 Preservation and packaging. Preservation and packaging which may be the supplier's commercial practice, shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.1.2 Packing. Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules or other carrier regulations as applicable to the mode of transportation and may conform to the suppliers commercial practice.

5.1.1.3 Marking. Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.1.2 Repair parts. Repair parts shall be preserved, packaged and packed for the level specified (see 6.2) in accordance with MIL-R-196. Repair parts not covered therein, shall be preserved in accordance with the guide lines of MIL-P-116. Repair parts shall be individually packaged unless used in sets or quantities greater than one.

5.2 Domestic shipment and storage or overseas shipment. The requirements and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.2).

6. NOTES

6.1 Intended use. Type I valves are intended for use in off and on service. Any throttling requirement should be very limited. Type II valves are specifically intended for severe throttling service, and should not be used for tight shut-off.

6.1.1 Class A, series 50. Class A, series 50 valves are intended primarily for low pressure sea water service.

6.1.2 Class A, series 200. Class A, series 200 valves are intended for fresh water or sea water service.

6.1.3 Class B, series 200. Class B, series 200 valves are intended primarily for JP-5 fuel service.

6.1.4 Class C, series 200. Class C, series 200 valves are intended primarily for fuel oil (Navy special) service.

6.1.5 Class D, series 200. Class D, series 200 valves are intended primarily for diesel or lube oil service.

6.1.6 Construction A. Construction "A" valves are intended for general use in sizes 2 to 20 inches.

6.1.6.1 Construction B. All sizes, may be furnished with lugs drilled only, or drilled and tapped to receive the flange bolts.

6.1.7 Installation.

(a) Ferrous body valves are for use with socket-weld or weld-neck, schedule 40, 150# flanges, in accordance with USAS B16.5.
(b) Non-ferrous body valves are normally intended for use with flanges detailed on Drawing 810-1385892.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type, class and series required (see 1.2).
(c) Materials required (see 3.2).
(d) Construction required (see 3.3).
(e) If means for positively locking disc in the fully open and fully closed position are to be provided (see 3.4.1).
(f) When valves in sizes 8 inch and above are to be provided with other than gear operators (see 3.4.1).
(g) Type of operation required, service application and environmental conditions which will effect the material selection in the valve operator (see 3.4.2).
(h) When remote operating devices are required (see 3.4.2.1).
(i) When construction "B" valve lugs are to be furnished drilled and tapped (see 3.4.3.2.2).
(j) When production drawings are required (see 3.5.2).
(k) Repair parts required (see 3.6.1).
(l) Levels of preservation, packaging and packing required for repair parts (see 5.1.2).
(m) Preservation, packaging and packing requirements if other than those required by 5.1 (see 5.2).

6.3 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids qualified for inclusion in applicable Qualified Products List QPL 22133 whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Ship Engineering Center, Department of the Navy, Center Building, Prince George's Center, Hyattsville, Maryland 20782, and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.3.1).

6.3.1 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

6.4 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in Section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.5 CHANGES FROM PREVIOUS ISSUE. THE OUTSIDE MARGINS OF THIS DOCUMENT HAVE BEEN MARKED "#" TO INDICATE WHERE CHANGES (DELETIONS, ADDITION, ETC.) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS HAS BEEN DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT AS WRITTEN IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

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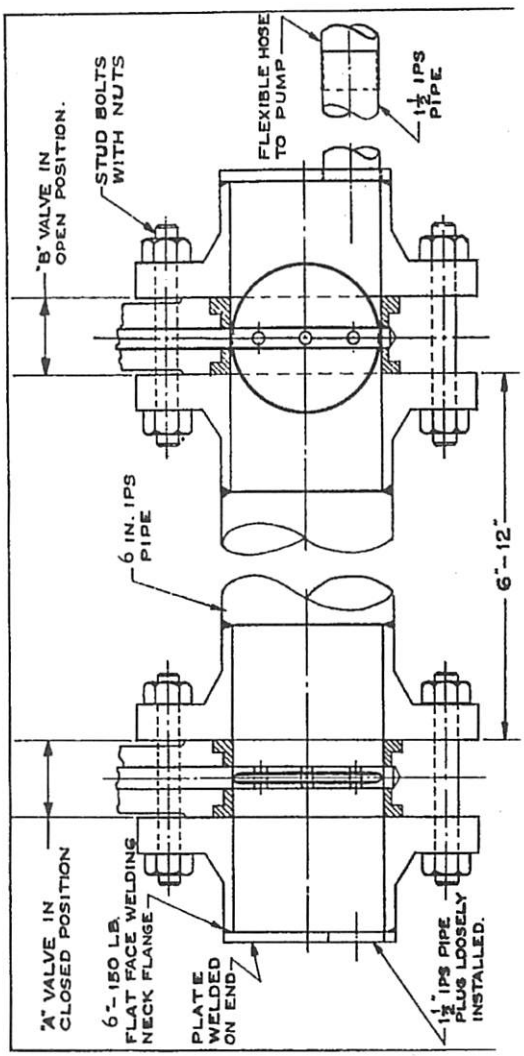


Figure 3 - Test assembly.

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